University of California, Riverside
Electrical and Computer Engineering Department

Electromagnetics I

Lecture 8: Maxwell's Equations in Material Region

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Maxwell's Equations in Materials

	INTEGRAL FORM	DIFFERENTIAL FORM
Gauss's law of electric field	$\oint_s \mathbf{D} \cdot ds = \int_v \rho_v \, dv$	$\nabla \cdot \mathbf{D} = \rho_v$
Gauss's law of magnetic field	$\oint_s \mathbf{B} \cdot ds = 0$	$\nabla \cdot \mathbf{B} = 0$

	INTEGRAL FORM	DIFFERENTIAL FORM
Faraday's law	$\oint_c \mathbf{E} \cdot d\ell = -\frac{d}{dt}\int_s \mathbf{B} \cdot ds$	$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$
Ampere's law	$\oint_c \mathbf{H} \cdot d\ell = \int_s \mathbf{J} \cdot ds + \int_s \sigma\mathbf{E} \cdot ds + \frac{d}{dt}\int_s \mathbf{D} \cdot ds$	$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t} + \sigma \mathbf{E}$

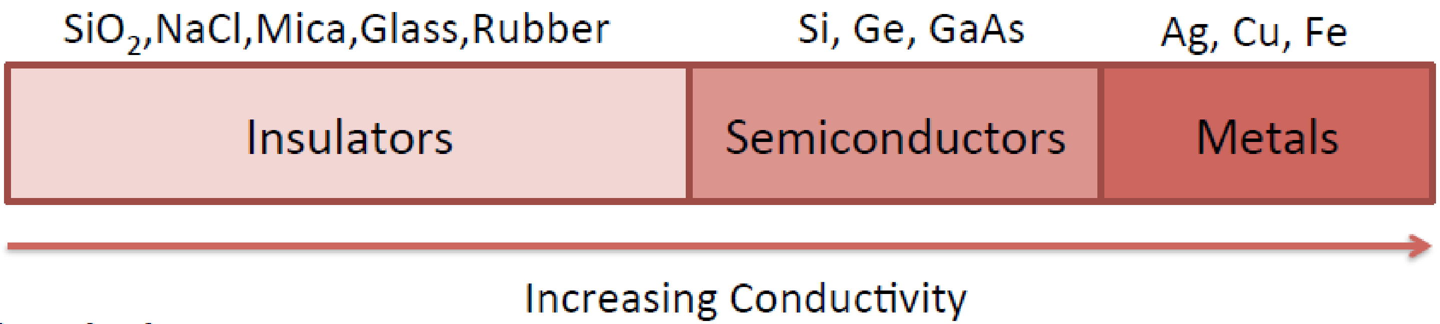
where $\mathbf{D} = \epsilon_o \epsilon_r \mathbf{E}$ and $\mathbf{B} = \mu_o \mu_r \mathbf{H}$

Outline

- Conductivity
- Dielectric Materials and Polarization
- Magnetic Materials and Magnetization
- Maxwell's Equations in Material Regions
- Boundary Conditions

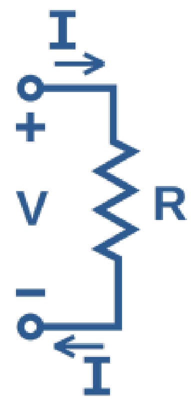
Chapter 3: 3.1 - 3.10

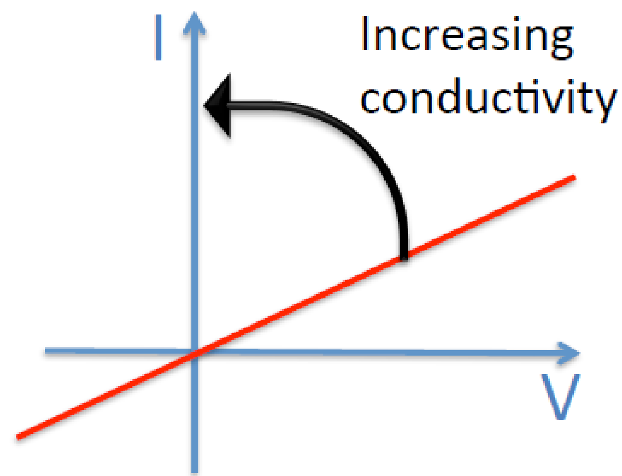
Ohm's Law and Conductivity



- Ohm's law:

$$\mathbf{R}\,[\Omega] = \frac{\mathbf{V}\,[volts]}{\mathbf{I}\,[amps]} = \textbf{constant}$$

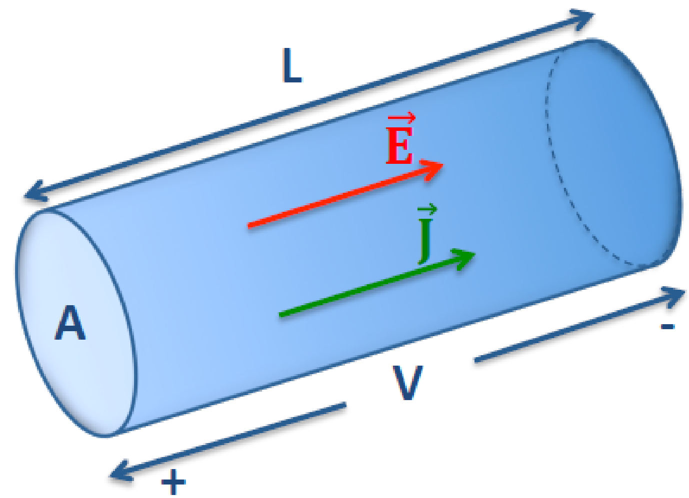




- Question: Is it always constant?
  R (Temperature)
  Non ohmic material: R(V), R(I) → Nonlinear I-V

Ohm's Law (Microscopic)

Resistance: $\mathbf{R}[\Omega] = \boldsymbol{\rho}[\,\Omega.m]\times\dfrac{\mathbf{L}\ [m]}{\mathbf{A}\ [m^2]}$



Current density: $\mathbf{J}\ \left[\dfrac{A}{m^2}\right] = \dfrac{\mathbf{I}\ [A]}{\mathbf{A}\ \ [m^2]}$

Electric field: $\mathbf{E}\ \left[\dfrac{V}{m}\right] = \dfrac{\mathbf{V}\ [volts]}{\mathbf{L}\ [m]}$

Inserting in ohm's law ➔ $\boxed{\boldsymbol{\rho} = \dfrac{\mathbf{E}}{\mathbf{J}}\quad \textbf{OR} \qquad J = \sigma E}$

Conductivity: $\boldsymbol{\sigma}\ \left[\dfrac{S}{m}\right] = \dfrac{\mathbf{1}}{\boldsymbol{\rho}}$

Advantages

1. Based on material characteristics, not dimension dependent
2. A vector relation ($\vec{\mathbf{E}}$ & $\vec{\mathbf{J}}$)
3. Valid at any local point

Resistivity, Resistance, Conductivity, and Conductance

- Resistance R Ohms [Ω]: is defined as the ratio of voltage across it (V) to current through it (I):

$$R = \frac{V}{I}$$

- Conductance G siemens (S): is the inverse:

$$G = \frac{I}{V}$$

- Resistivity ρ ohm-metres (Ω·m) is a measure of the material's ability to oppose electric current:

$$R = \rho \frac{L}{A}$$

- Conductivity σ [S/m] is a measure of the material's ability to oppose electric current:

$$G = \sigma \frac{A}{L}$$

Physical Essence

- Conductivity and resistivity are the material parameters
- Resistance and conductance depends on the material parameters as well as on the sample shape

$$R = \rho \frac{L}{A}$$

$$G = \sigma \frac{A}{L}$$

L – sample length (m)
A – sample cross section (m^2)

Ohm's Law

- Georg Simon Ohm was born in Erlangen, Germany, in 1787 and entered that city's university in 1805, where he received a doctorate. He taught mathematics in local schools and performed experiments in physics in a school physics laboratory, trying to master the principles behind electromagnetism. He became professor of mathematics at the Jesuits' College in Cologne in 1817.

- In 1826, he published papers giving a mathematical model for the way the circuits conducted heat in Fourier's studies. In May 1827, Ohm published Die galvanische Kette, mathematisch bearbeitet, which described the relationship between electromotive force, current, and resistance later known as Ohm's law. Ohm obtained the experimental data from which he first formulated his law on 8 January 1826. But his study received a muted reception upon its initial release, and he resigned his position at Cologne, eventually taking a new professorship in Nurenberg in 1833.

- Ohm's findings would catalyze new research into electricity in the coming decades. In 1841, Ohm was awarded the Royal Society's highest award, the Copley Medal. The term 'Ohm' was tagged as the unit of electrical resistance in 1872.

Georg Simon Ohm

Siemens (unit)

- The siemens (symbol: S) is the derived unit of electric conductance, electric susceptance, and electric admittance in the International System of Units (SI). Conductance, susceptance, and admittance are the reciprocals of resistance, reactance, and impedance respectively; hence one siemens is redundantly equal to the reciprocal of one ohm, and is also referred to as the mho. The 14th General Conference on Weights and Measures approved the addition of the siemens as a derived unit in 1971.
- The unit is named after Ernst Werner von Siemens. In English, the same form siemens is used both for the singular and plural.

13 December 1816 – 6 December 1892) was a German inventor and industrialist. Siemens's name has been adopted as the SI unit of electrical conductance, the siemens. He was also the founder of the electrical and telecommunications company Siemens.

Apart from the pointer telegraph Siemens made several contributions to the development of electrical engineering and is therefore known as the founding father of the discipline in Germany. He built the world's first electric elevator in 1880.[8] His company produced the tubes with which Wilhelm Conrad Röntgen investigated x-rays. He claimed invention of the dynamo although others invented it earlier. On 14 December 1877 he received German patent No. 2355 for an electromechanical "dynamic" or moving-coil transducer, which was adapted by A. L. Thuras and E. C. Wente for the Bell System in the late 1920s for use as a loudspeaker.[9] Wente's adaptation was issued US patent 1,707,545 in 1929. Siemens is also the father of the trolleybus which he initially tried and tested with his "Elektromote" on 29 April 1882.

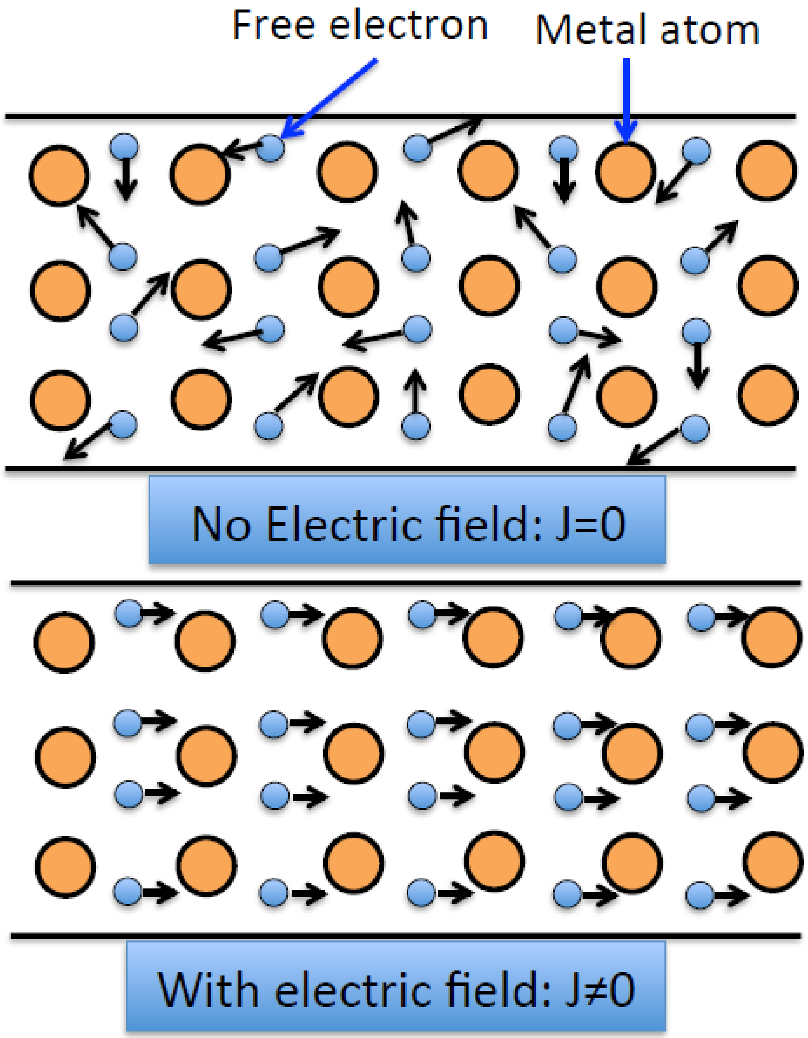
Invented by Dr. Ernst Werner Siemens

The "Elektromote", the world's first trolleybus in Berlin, Germany, 1882

Trolleybus in Cambridge, Massachusetts.

Drude’s Classical Free Electron Theory

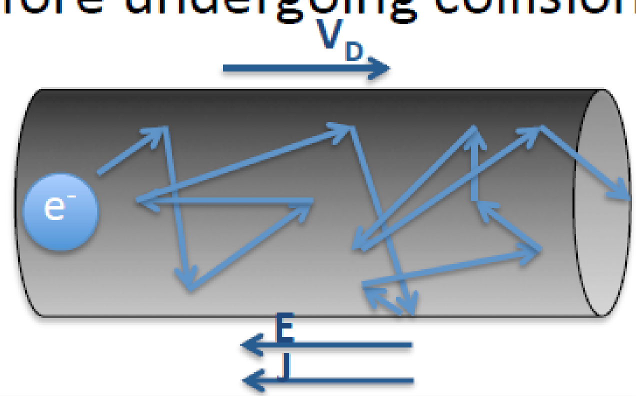
Paul Drude
German physicist
(1863-1906)



- ✓ fixed lattice and “free” electron gas
- ✓ electrons are treated as charged particles
- ✓ random collisions by other electrons, boundaries, lattice

Continue with Drude's Model

- **Drift Velocity (V_D):** The average velocity acquired by electrons in particular direction under the application of electric field.
- **Mobility (μ):** Drift velocity normalized by electric field.
- **Relaxation time(τ):** The average time between collisions.
- **Mean Free Path (MFP):** Average distance an electron travel before undergoing collision (MFP = $\tau \times V_{th}$)



Eq. Box (1-1)

$$\boxed{\sigma = e^- N_e \mu_e}$$

$$\boxed{\mu_e = e^- \tau m_e}$$

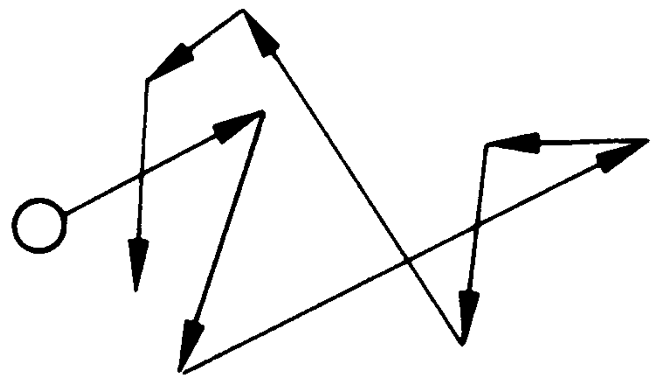
Conductivity is defined based on charge career density and their mobility

examples

Conductivity and Mobility

Equilibrium

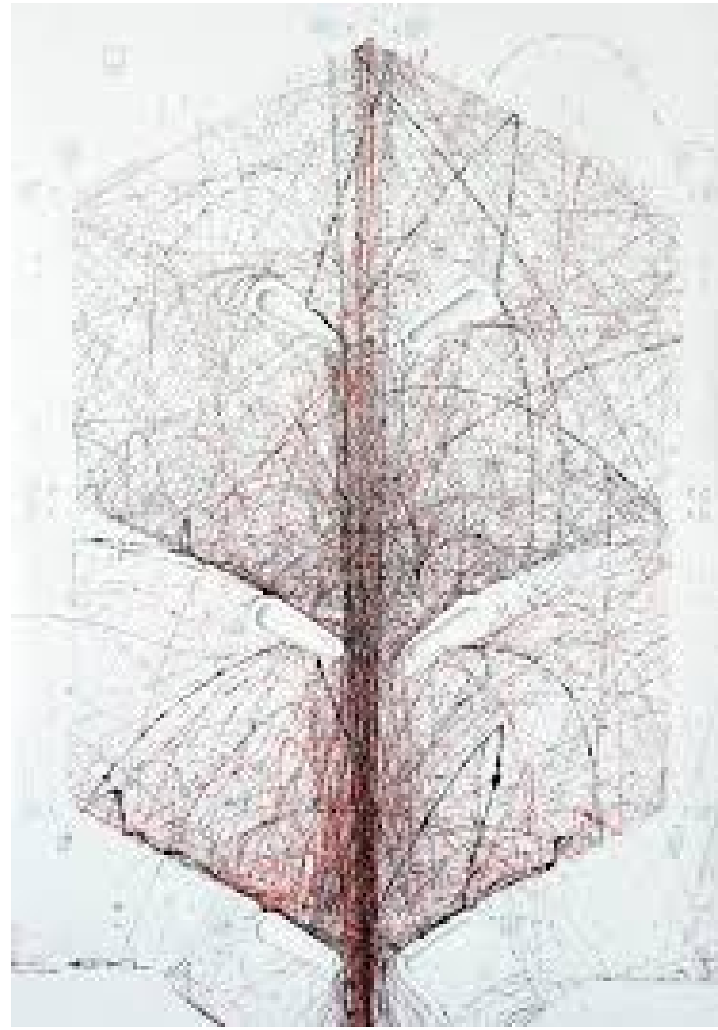
- Thermal (random) motion of electrons and holes
- The average of the velocity vector <$\mathbf{v}_{th}$> = 0.



- The average kinetic energy is $\frac{1}{2}m^* v_{th}^2 = \frac{3}{2}k_B T$

where m* is the conductivity effective mass.

Pinball Machine

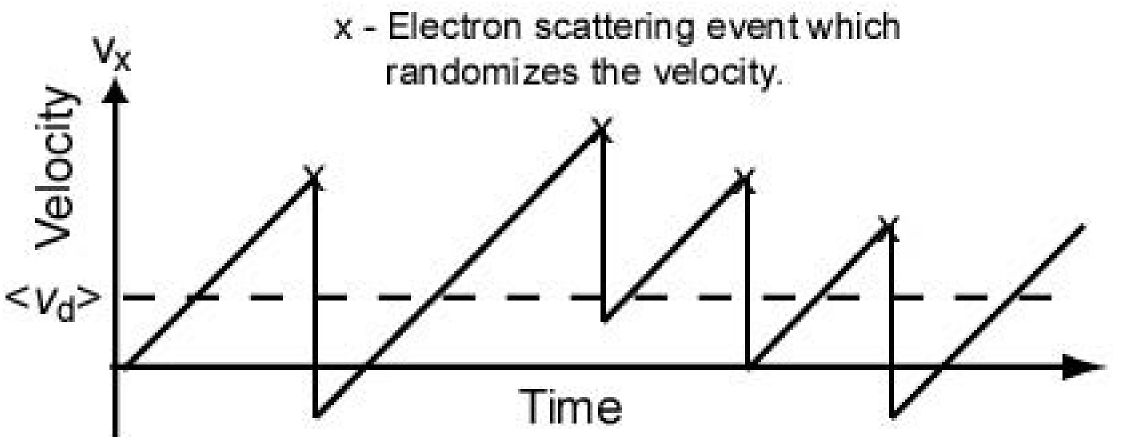


Mobility

Consider one electron and apply an ***E*** field.

- $\boldsymbol{F} = -q\mathcal{E} = m^*\boldsymbol{a}$
- $\boldsymbol{v}_d(t) = \boldsymbol{a}\, t = -(q\mathcal{E}/m^*)\, t$ (free flight)

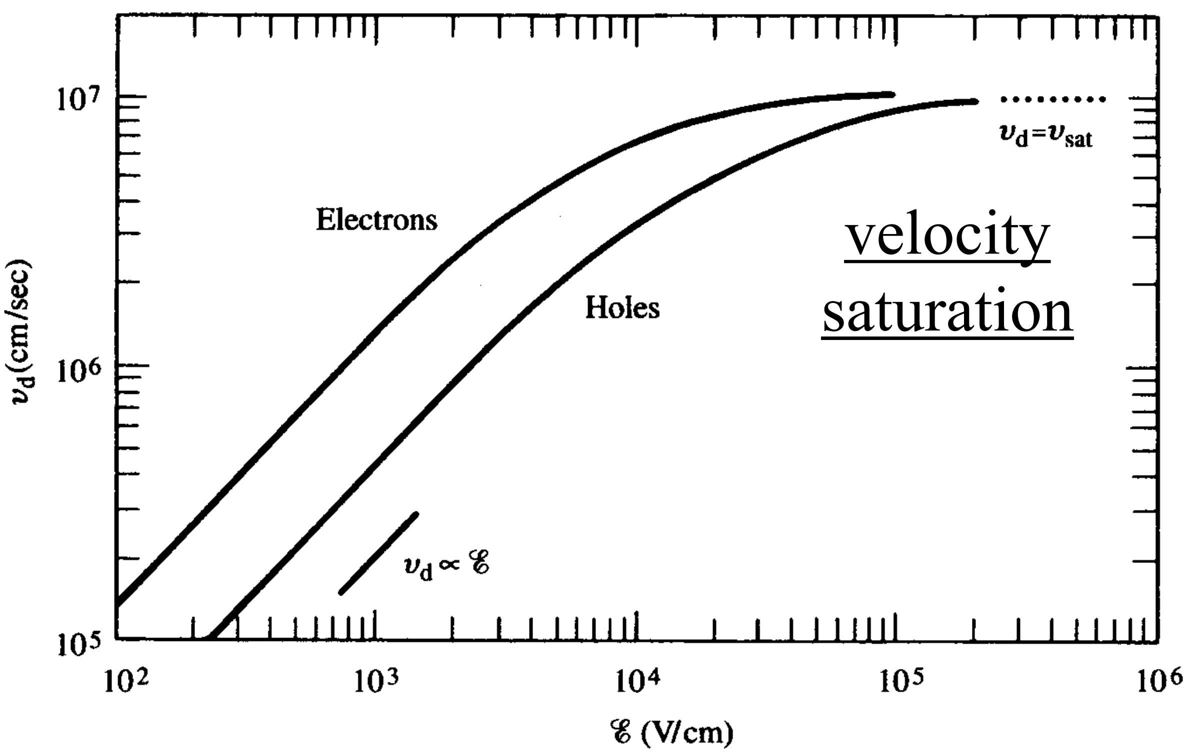
$$\mathcal{E} = -|\mathcal{E}|\hat{x}$$



Average velocity: $\langle \boldsymbol{v}_d \rangle = \frac{1}{T}\int_0^T dt\, \boldsymbol{v}(t)$

High Electric Fields: Velocity Saturation

- Low E-fields: $\langle \boldsymbol{v}_n \rangle \equiv -\mu_n \boldsymbol{\mathcal{E}}$ proportional to $\boldsymbol{\mathcal{E}}$.
- High E-fields: $\left|\langle \boldsymbol{v}_n \rangle\right| \equiv v_{\text{sat}}$ = saturation velocity



$v_{\text{sat}} \approx 10^7$ cm/s

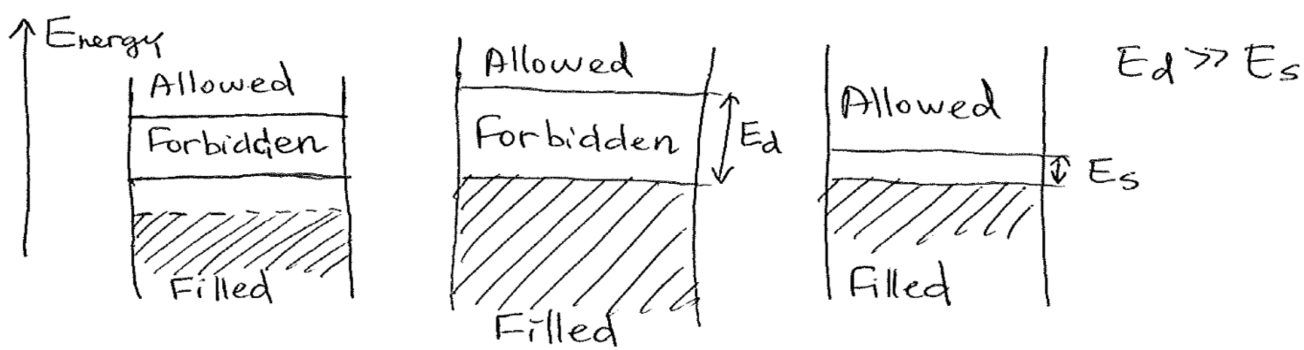
Modern FETs operate in this high field regime.

Conductors and Semiconductors

$\vec{J} = \sigma \vec{E}$ (A/m^2); σ – conductivity (S/m)

Material	Conductivity, σ (S/m)	
Silver	$6.2 \cdot 10^7$	conductors
Copper	$5.8 \cdot 10^7$	conductors
Iron	10^7	conductors
Ge (pure)	2.2	semiconductors
Si (pure)	$4.4 \cdot 10^{-4}$	semiconductors
Glass	10^{-12}	insulators
Fused quartz	10^{-17}	insulators

Band Diagrams



Conductor Dielectric Semiconductor

Conduction Current Density

The quantity of current or charges that pass across the conduction surface in time t is referred to as the conduction current density. The conduction surface is parallel to the current flow in this case. Conduction current density formula

$$\overrightarrow{J_c} = \sigma \vec{E}$$

Conduction current J_c appears in conductors due to the external electric field.

Units J_c: [A/m^2]

Anisotropy in Electrical Conduction

Isotropic:

similar in-plane and cross-plane elements

Cubic crystals, randomly oriented polycrystals

Anisotropic:

different in-plane and cross-plane elements

Single-crystals, polycrystals of preferred orientat

In General in three space axis:

$$[J] = [\sigma]_{3\times3}\,[E]$$

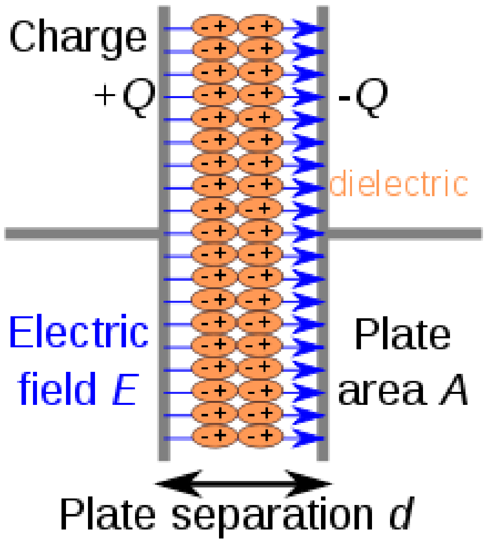
$$J_1 = \sigma_{11} E_1 + \sigma_{12} E_2 + \sigma_{13} E_3$$
$$J_2 = \sigma_{21} E_1 + \sigma_{22} E_2 + \sigma_{23} E_3$$
$$J_3 = \sigma_{31} E_1 + \sigma_{32} E_2 + \sigma_{33} E_3$$

Electrical Resistivity (ohm-m)	(log)		Electrical Conductivity (S/m)
	−8	8	copper, silver
intercalated graphite	−7	7	iron
	−6	6	stainless steel, metallic glass
graphite (in-plane) graphite (out of plane)	−5	5	$YBa_2Cu_3O_7$ (ab plane)
polyacetylene (doped)	−4	4	$YBa_2Cu_3O_7$ (c-axis)
	−3	3	silicon (doped) ZnO (doped)
TTF-TCNQ	−2	2	seawater
	−1	1	Fe_3O_4
	0	0	germanium
	1	−1	
	2	−2	
	3	−3	silicon
	4	−4	InSb
	5	−5	water AgCl
polyacetylene (undoped)	6	−6	
	7	−7	
	8	−8	ZnO (undoped)
	9	−9	
	10	−10	
Bakelite	11	−11	NaCl
polypyrrole	12	−12	
	13	−13	
Lucite (PMMA)	14	−14	Al_2O_3
polyvinyl chloride	15	−15	mica silica
polyethylene, teflon	16	−16	diamond

Ref. Livingston, James D. Electronic properties of engineering materials. New York: Wiley, 1999.

Dielectrics

- In electromagnetism, a dielectric (or dielectric medium) is an electrical insulator that can be polarised by an applied electric field. When a dielectric material is placed in an electric field, electric charges do not flow through the material as they do in an electrical conductor, because they have no loosely bound, or free, electrons that may drift through the material, but instead they shift, only slightly, from their average equilibrium positions, causing dielectric polarisation. [Wiki]



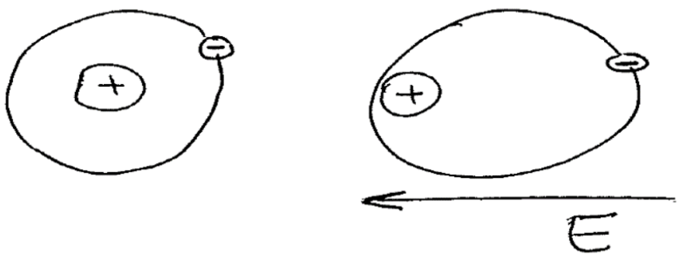
Dielectric Materials and Polarization

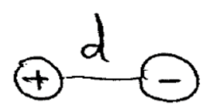
Dielectrics: bound electrons are predominant

E : displacement of bound charges, not drift.

Different types of polarization

Electronic





$\vec{p} = Q \cdot \vec{d}$

electric dipole

Orientational polarization

polar dielectric materials

Ionic Polarization

NaCl (sodium chloride)

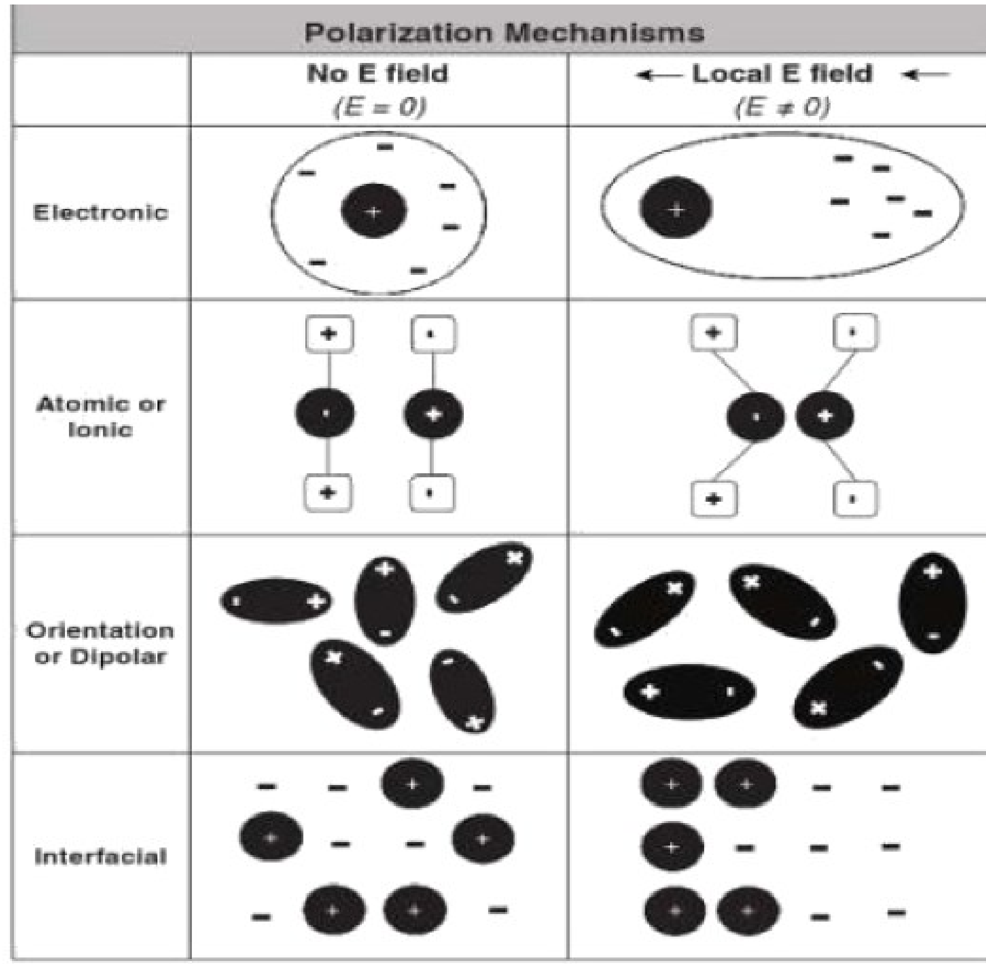
Types of polarization mechanisms

Electronic Polarization: This effect occurs in all atoms under the application of an electric field. The nucleus of the atom and the center of its electron cloud shift away from each other, creating a tiny dipole with very small polarization effect.

Ionic Polarization: In ionic solids such as ceramic materials, the ions are symmetrically arranged in a crystal lattice with a net zero polarization. Once an electric field is applied, the cations and anions are attracted to opposite directions. This creates a relatively large ionic displacement (compared to electronic displacement), which can give rise to high dielectric constants in ceramics popularly used in capacitors.

Dipole (or Orientation) Polarization: Certain solids have permanent molecular dipoles that, under an electric field, rotate themselves in the direction of the applied field, creating a net average dipole moment per molecule. Dipole orientation is more common in polymers since their atomic structure permits reorientation.

Space Charge (or Interfacial) Polarization: In ceramics, this phenomenon arises from extraneous charges that come from contaminants or irregular geometry in the interfaces of the polycrystalline solids. These charges are partly mobile and migrate under an applied field, causing this extrinsic type of polarization.



https://blog.knowlescapacitors.com/blog/capacitor-fundamentals-part-4-dielectric-polarization

Polarization

electric dipole moment $\vec{p} = Q\vec{d}$ – microscopic

$$\boxed{\vec{P} = \lim_{\Delta V \to 0} \frac{1}{\Delta V} \sum_{i=1}^{N} \vec{p}_i = n \cdot \vec{p}_a = n Q \vec{d}_a}$$

$N = n \cdot \Delta V$ – number of dipoles in a volume ΔV.

$\vec{p}_a$ – average dipole moment.

Polarization $\vec{P}$ – macroscopic characteristic of the material.

In most materials $\vec{P}$ is linearly proportional to the applied field

$$\vec{P} = \epsilon_0 \chi_e \vec{E}$$

χ_e – electric susceptibility (tensor in general case)

Polarization Current:

$$\vec{J}_p = \frac{\partial \vec{P}}{\partial t} = \frac{\partial (\epsilon_0 \chi_e \vec{E})}{\partial t}$$

$$\boxed{\epsilon_r = 1 + \chi_e}$$

electric flux density: $\vec{D} = \epsilon_0 \epsilon_r \vec{E}$.

Dielectric Constant

(Relative permittivity)

Dielectric Constant is the ratio between the permittivity of the medium to the permittivity of free space.

Absolute permittivity (of material)

Relative permittivity $$\varepsilon_r = \frac{\varepsilon}{\varepsilon_0}$$

Permittivity of vacuum

its value changes widely from material to material

For vacuum =1

For all other dielectric it is $\varepsilon r > 1$.

So, we can write $\varepsilon r = 1 + \chi e$, χe is susceptibility

The characteristics of a dielectric material are determined by the dielectric constant and it has no units.

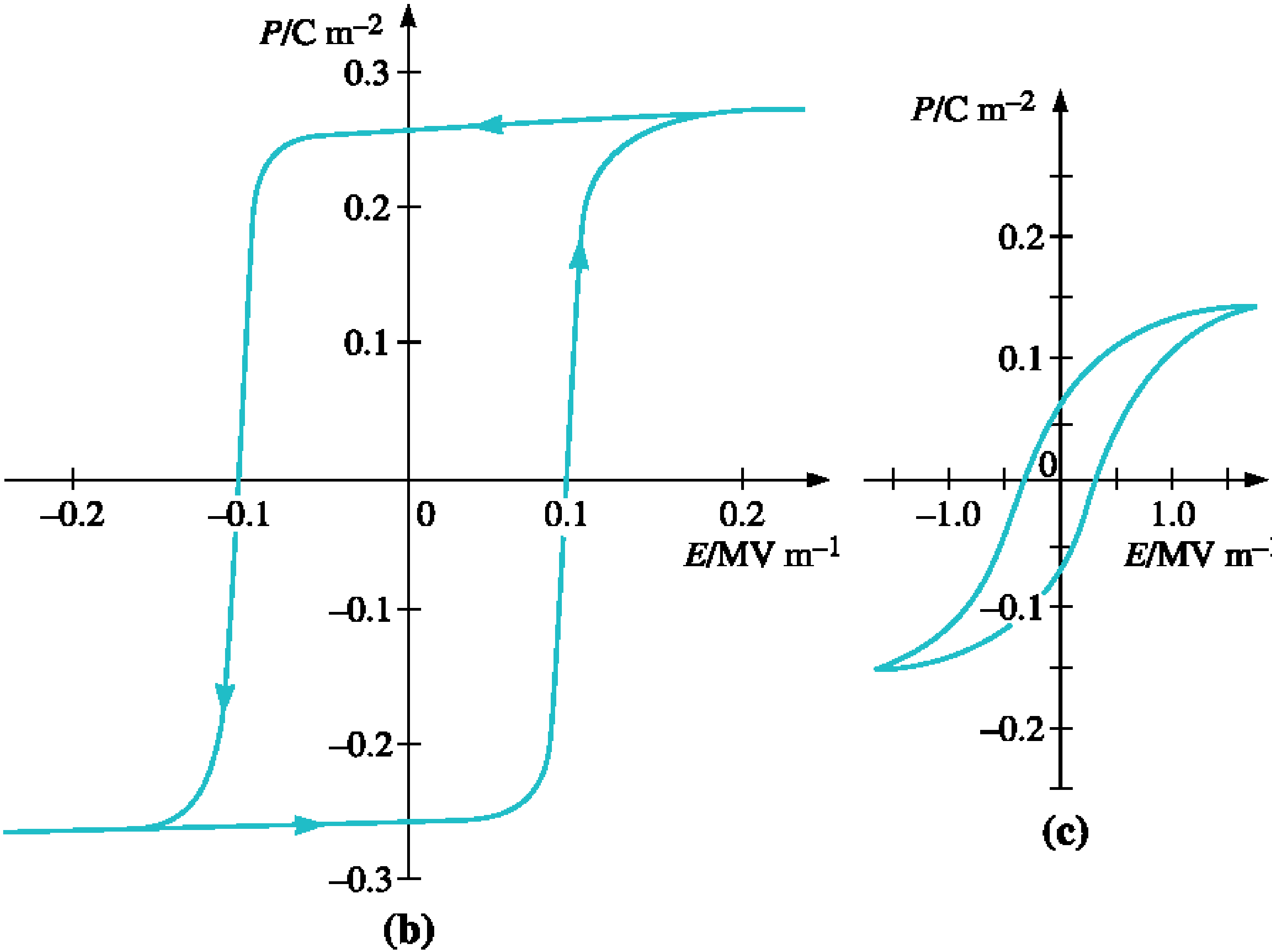
Dielectric constant of different materials

Dielectric Material	Dielectric Constant
Vacuum	1
Air	1.006
Polypropylene PP	2.2
Polyphenylene Sulfide PPS	3
Polyester PET	3.3
Polyester PEN	3
Impregnated Paper	2.0 – 6.0
Mica	6.8
Aluminium Oxide	8.5
Tantalum Oxide	27.7
Paraelectric Ceramics (Class1)	5.0 – 9.0
Barium Titanate (Class 2)	3000 - 8000

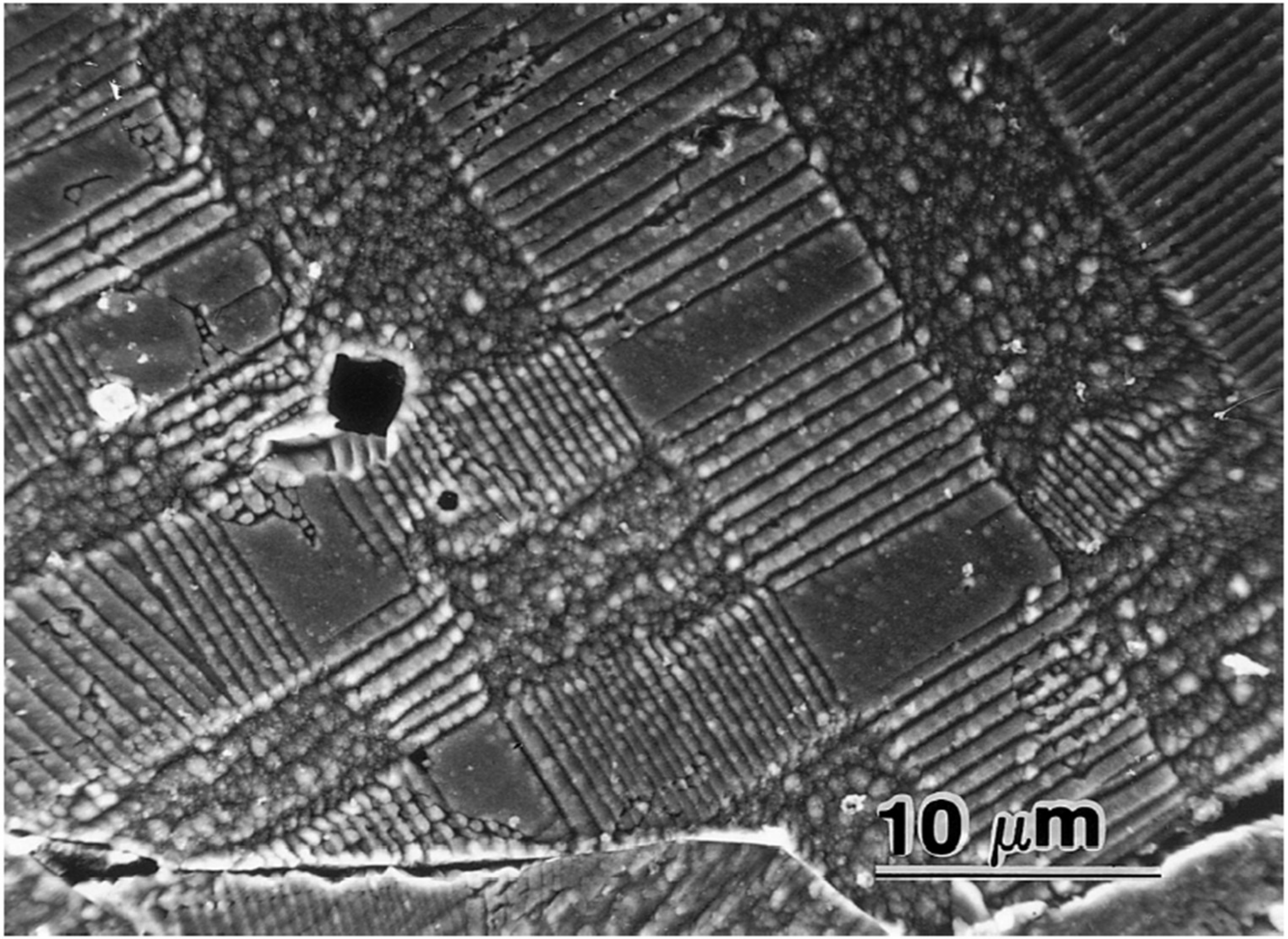
Types of Dielectrics

- Ferroelectric - A material that shows spontaneous and reversible dielectric polarization.
- Piezoelectric – A material that develops voltage upon the application of a stress and develops strain when an electric field is applied.

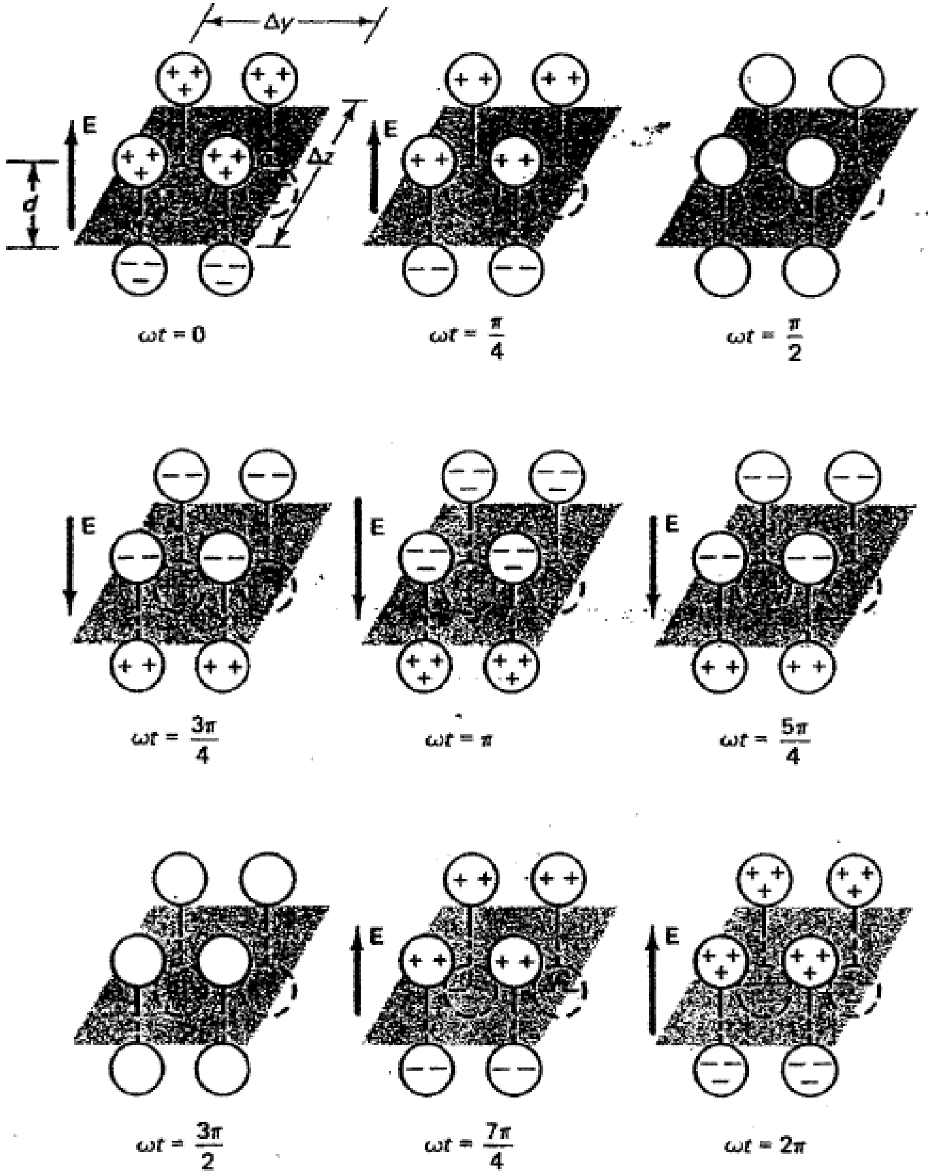
Polycrystalline BaTiO3



Ferroelectric domains in polycrystalline BaTiO3.



Polarization Current Density



$$\boxed{\begin{aligned}\mathbf{J}_p &= \frac{\partial \mathbf{P}}{\partial t} \\ &= \frac{\partial(\epsilon_o \chi_e \mathbf{E})}{\partial t}\end{aligned}}$$

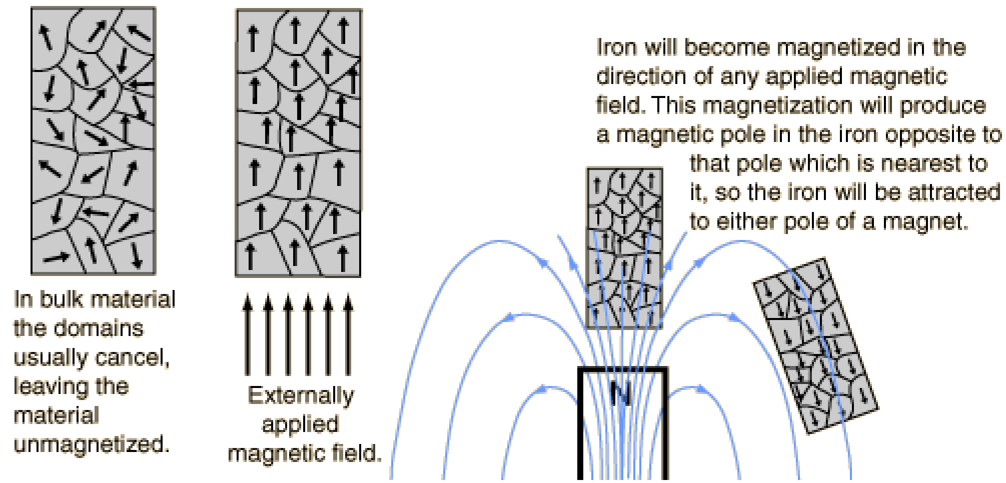
Polarization current density $J_p =$ is equal to the rate of change of polarization

Units J_p: [A/m^2]

Magnetic Field in Material

$\vec{M}$ is the total magnetic moment per unit volume

$$\boxed{\vec{M} = \lim_{\Delta v \to \infty} \frac{1}{\Delta v} \sum_{i=1}^{n \Delta v} \overrightarrow{m_i}}$$



$$\vec{M} = \chi_m \cdot \vec{H}$$

χ_m is the magnetic susceptibility of the material (dimensionless)

$$\mu_r = (1 + \chi_m)$$

$$\vec{B} = \mu_0 (1 + \chi_m) \vec{H} = \mu_0 \cdot \mu_r \cdot \vec{H}$$

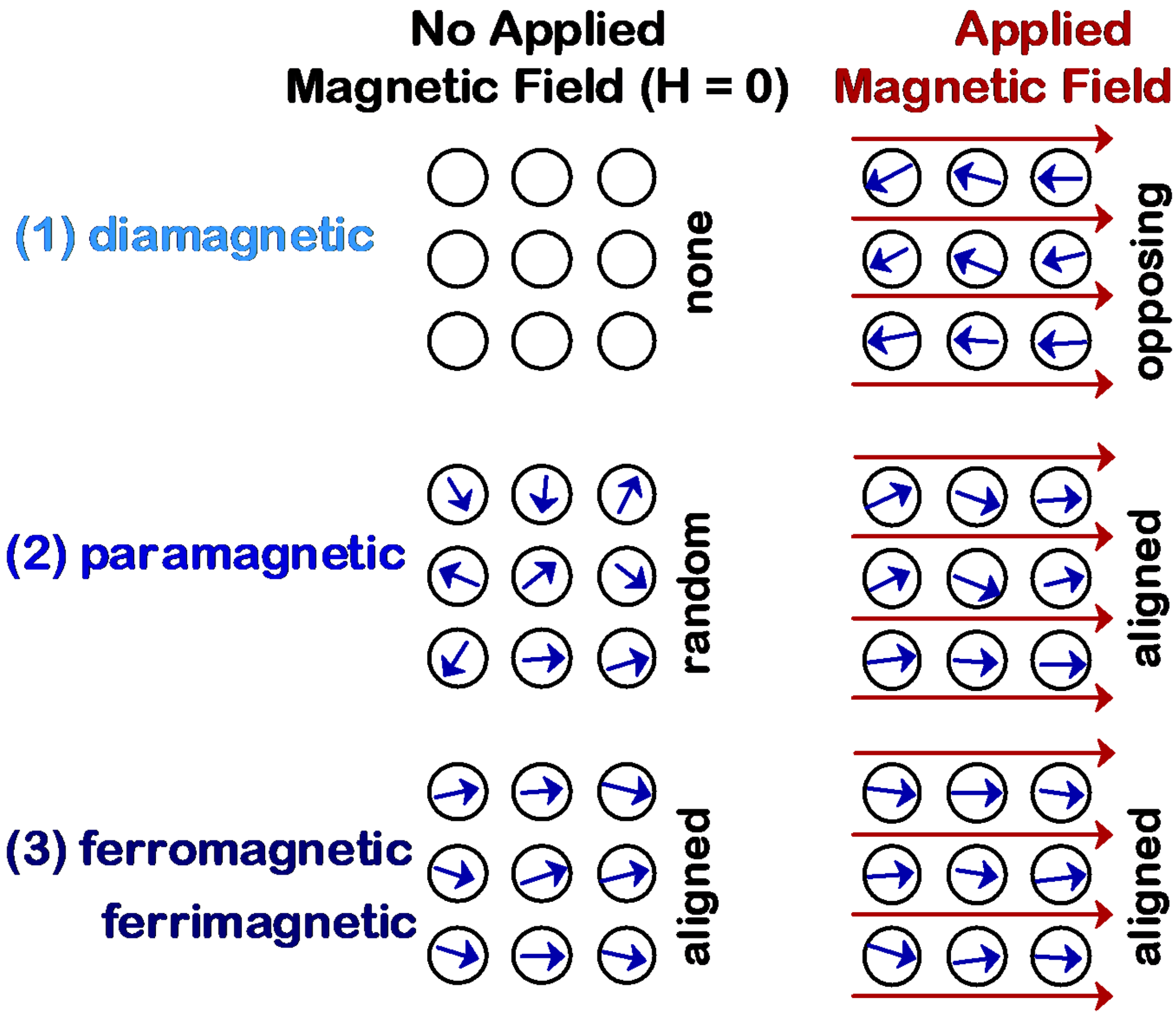
Magnetic Properties of Materials

Material	χ_m [cm^3/g]	χ_v unitless	μ unitless	Type of magnetism
Bi	-1.34×10^{-6}	-13.13×10^{-6}	0.99983	
Be	-1.0×10^{-6}	-1.85×10^{-6}	0.99998	
Ag	-0.192×10^{-6}	-2.016×10^{-6}	0.99997	Diamagnetic
Au	-0.142×10^{-6}	-2.74×10^{-6}	0.99996	
Ge	-0.106×10^{-6}	-0.564×10^{-6}	0.99999	
Cu	-0.086×10^{-6}	-0.77×10^{-6}	0.99999	
Superconductors[a]		$\sim -8 \times 10^{-2}$		
Snβ	$+0.026 \times 10^{-6}$	$+0.19 \times 10^{-6}$	1	
W	$+0.32 \times 10^{-6}$	$+6.18 \times 10^{-6}$	1.00008	
Al	$+0.61 \times 10^{-6}$	$+1.65 \times 10^{-6}$	1.00002	Paramagnetic
Pt	$+0.983 \times 10^{-6}$	$+21.04 \times 10^{-6}$	1.00026	
Mn	$+8.9 \times 10^{-6}$	$+66.13 \times 10^{-6}$	1.00083	
Low carbon steel			5×10^3	
Fe–3% Si (grain oriented)			4×10^4	Ferromagnetic
Ni–Fe–Mo			10^6	

[a] See Section 7.6.1.
Note: The table lists the mass susceptibility (or specific susceptibility), χ_m, whose unit is cm^3/g and the unitless volume susceptibility, χ_v, which was calculated from χ_m by multiplying χ_m by the density. Other sources may provide mass, atomic, molar, or gram equivalent susceptibilities in cgs or mks units.

... plus antiferromagnetic and ferrimagnetic

MAGNETIC MOMENTS FOR 3 TYPES



Adapted from Fig. 20.5(a), *Callister 6e*.

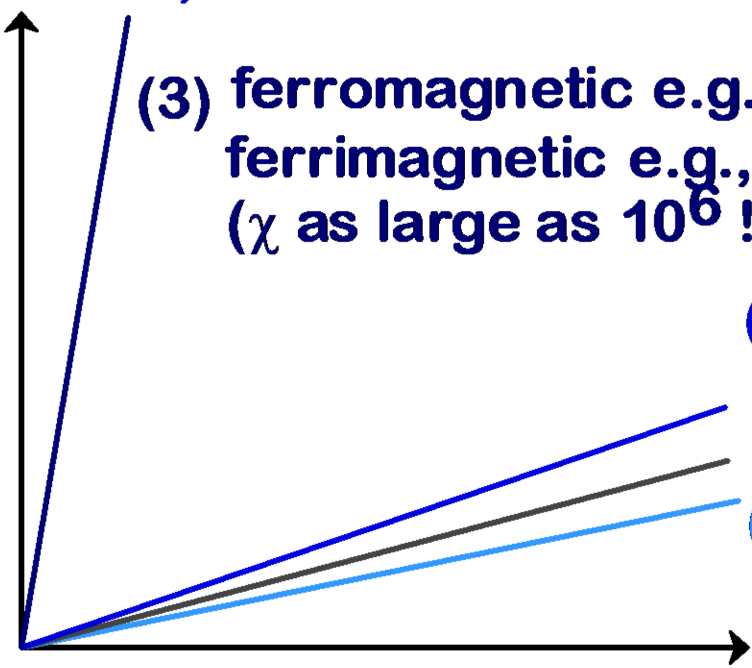
Adapted from Fig. 20.5(b), *Callister 6e*.

Adapted from Fig. 20.7, *Callister 6e*.

3 TYPES OF MAGNETISM

$$B = (1 + \chi)\mu_o H$$

permeability of a vacuum:
(1.26 x 10^{-6} Henries/m)



Plot adapted from Fig. 20.6, *Callister 6e*. Values and materials from Table 20.2 and discussion in Section 20.4, *Callister 6e*.

Ig Nobel Prize in Physics

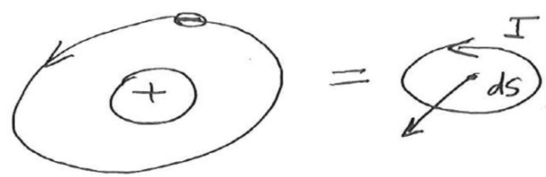
Improbable research is research that makes people laugh and then think

Geim and Berry Ig Nobel Prize in

Physics 2000:Frog diamagnetic levitation

Magnetic Materials and Their Magnetization

magnetization: alignment of the atomic magnetic dipoles along the direction of $\vec{H}$



$$\boxed{\vec{m} = I\, d\vec{s}}$$

magnetic dipole moment

also: electron & nucleus spins:

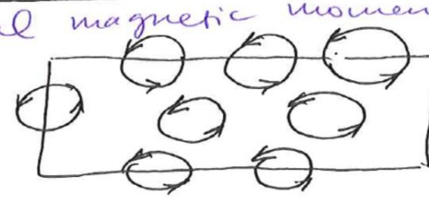
Magnetization:

$$\boxed{\vec{M} = \lim_{\Delta V \to 0} \frac{1}{\Delta V} \sum_{i=1}^{n\Delta V} \vec{m}_i = n\vec{m}_a = nI\,d\vec{s}}$$

— total magnetic moment per unit volume.

n - number of magnetic dipoles per unit volume

$\vec{m}_a$ - average mag. dipole moment

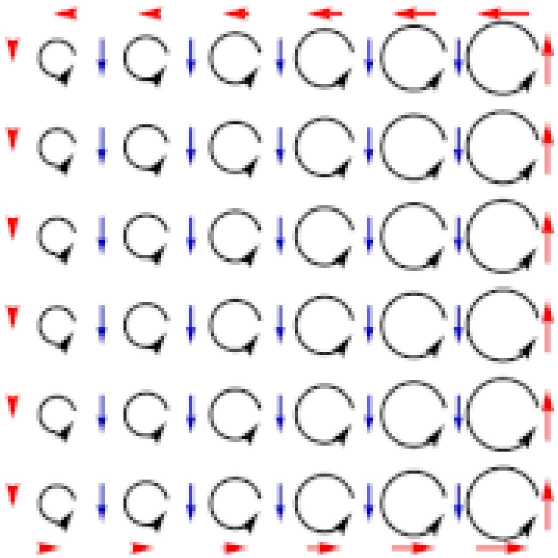


Magnetization current density:

$$\boxed{\vec{J}_m = \vec{\nabla} \times \vec{M}}$$

Ampere's Law: $\vec{\nabla} \times \dfrac{\vec{B}}{\mu_0} = \vec{J} + \dfrac{\partial \vec{D}}{\partial t} + \vec{J}_m$

Magnetization Current



When the microscopic currents induced by the magnetization (black arrows) do not balance out, bound volume currents (blue arrows) and bound surface currents (red arrows) appear in the medium. [Wiki]

$$\mathbf{J}_{\mathrm{m}} = \nabla \times \mathbf{M}$$

Units J_m: [A/m^2]

Gauss's Law for Electric Field in Materials

$$\nabla \cdot (\epsilon_o \mathbf{E} + \mathbf{P}) = \rho_v$$

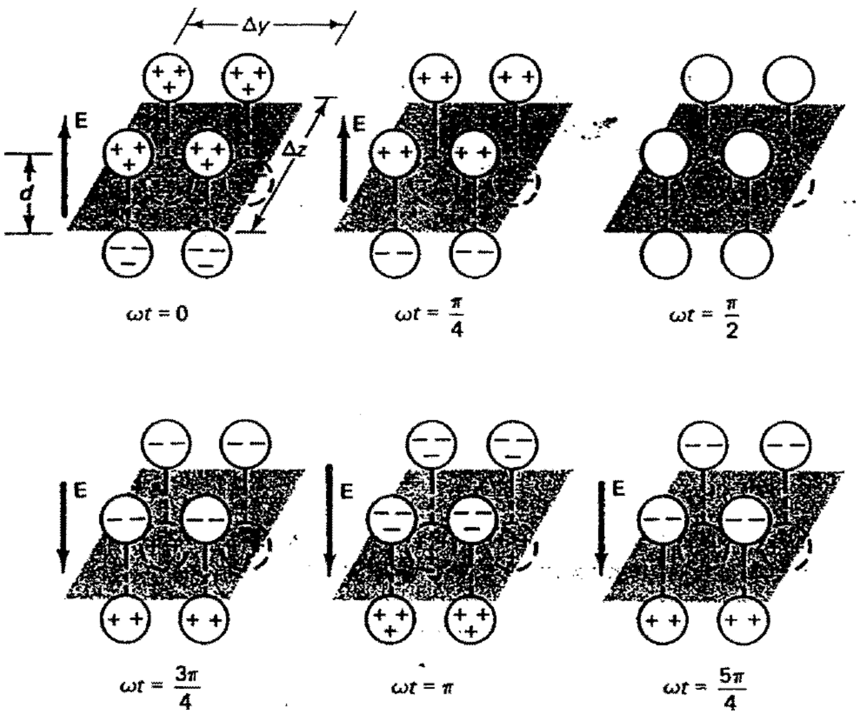
$$\mathbf{P} = \chi_e \epsilon_o \mathbf{E}$$

$$\nabla \cdot (\epsilon_o \mathbf{E} + \chi_e \epsilon_o \mathbf{E}) = \rho_v$$

$$\nabla \cdot \epsilon_o \mathbf{E}(1 + \chi_e) = \rho_v$$

$$\nabla \cdot \mathbf{D} = \rho_v$$

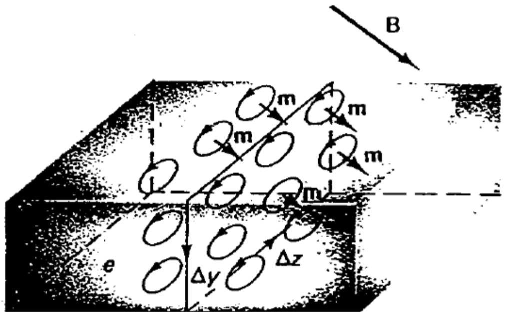
$$\boxed{\mathbf{D} = \epsilon_o(1 + \chi_e)\mathbf{E} = \epsilon_o \epsilon_r \mathbf{E}}$$

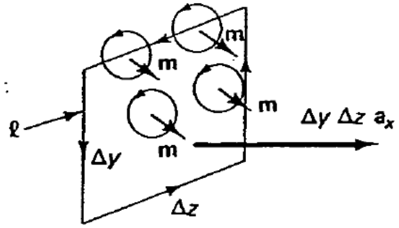


$$\boxed{\nabla \cdot \mathbf{D} = \rho_v}$$

$$\boxed{\oint_s \mathbf{D} \cdot d\mathbf{s} = \int_v \rho_v \, dv}$$

Gauss's Law for Magnetic Field in Materials



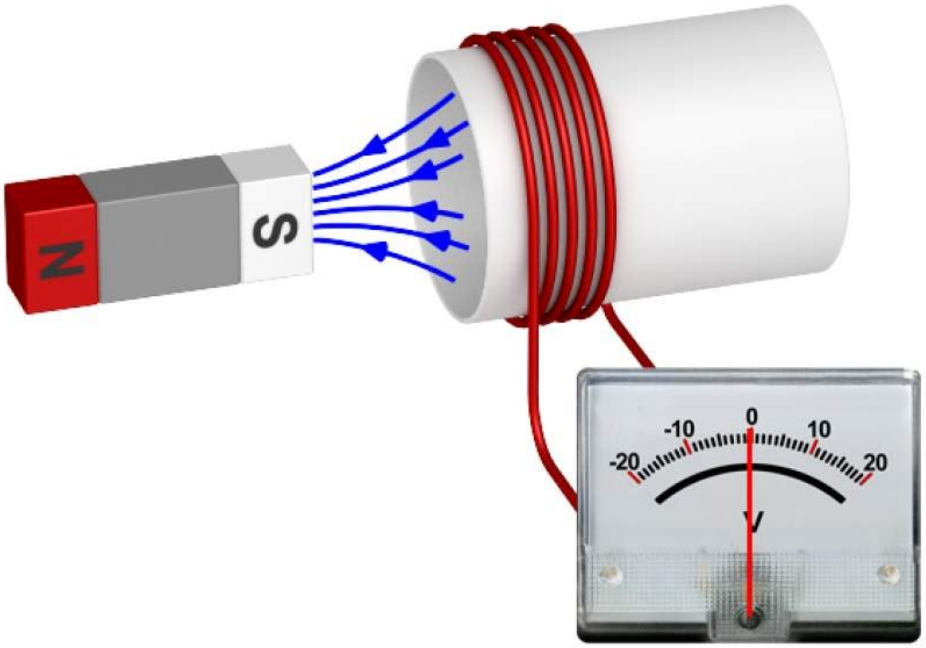


$$\boxed{\nabla \cdot \mathbf{B} = 0 \qquad \oint_s \mathbf{B} \cdot d\mathbf{s} = 0}$$

Faraday's Law in Materials

$$\boxed{\oint_c \mathbf{E} \cdot d\ell = -\frac{d}{dt}\int_s \mathbf{B} \cdot ds}$$

$$\boxed{\mathbf{B} = \mu_o \, \mu_r \, \mathbf{H}}$$



Ampere's Law in Materials

$$\boxed{\nabla \times \frac{\mathbf{B}}{\mu_o} = \mathbf{J} + \mathbf{J}_c + \mathbf{J}_m + \mathbf{J}_p + \frac{\partial \epsilon_o \mathbf{E}}{\partial t}}$$

On substituting $\mathbf{J}_c = \sigma \mathbf{E}$,

$$\mathbf{J}_m = \nabla \times \mathbf{M} = \nabla \times \chi_m \mathbf{H} \qquad \text{(for linear magnetic material)}$$

and

$$\mathbf{J}_p = \frac{\partial \mathbf{P}}{\partial t} = \frac{\partial \epsilon_o \chi_e \mathbf{E}}{\partial t} \qquad \text{(for linear dielectric material)}$$

$$\mathbf{H} = \frac{\mathbf{B}}{\mu_o} - \mathbf{M} = \frac{\mathbf{B}}{\mu_o} - \chi_m \mathbf{H}$$

$$\boxed{\nabla \times \mathbf{H} = \mathbf{J} + \sigma \mathbf{E} + \frac{\partial \epsilon \mathbf{E}}{\partial t}}$$

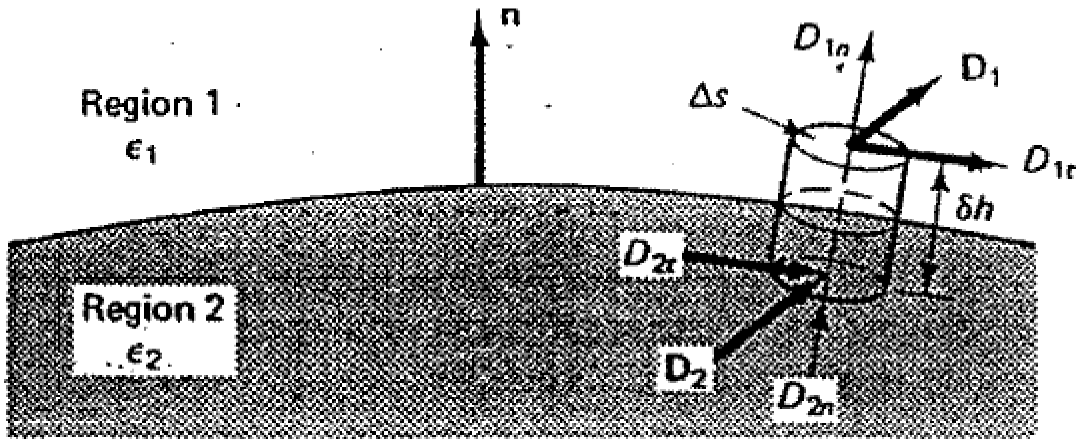
$$\boxed{\oint_c \mathbf{H} \cdot d\ell = \int_s \mathbf{J} \cdot d\mathbf{s} + \int_s \sigma \mathbf{E} \cdot d\mathbf{s} + \frac{d}{dt} \int_s \epsilon \mathbf{E} \cdot d\mathbf{s}}$$

Boundary Conditions

INTEGRAL FORMS	DIFFERENTIAL FORMS	
$\oint_s \epsilon_o \epsilon_r \mathbf{E} \cdot d\mathbf{s} = \int_v \rho_v dv$	$\nabla \cdot \epsilon_o \epsilon_r \mathbf{E} = \rho_v$	(4.1)
$\oint_s \mathbf{B} \cdot d\mathbf{s} = 0$	$\nabla \cdot \mathbf{B} = 0$	(4.2)
$\oint_c \mathbf{E} \cdot d\boldsymbol{\ell} = -\frac{d}{dt}\int_s \mathbf{B} \cdot d\mathbf{s}$	$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$	(4.3)
$\oint_c \mathbf{H} \cdot d\boldsymbol{\ell} = \int_s \mathbf{J} \cdot d\mathbf{s} + \frac{d}{dt}\int_s \epsilon_o \epsilon_r \mathbf{E} \cdot d\mathbf{s}$	$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \epsilon_o \epsilon_r \mathbf{E}}{\partial t}$	(4.4)

Normal Components of Electric Field

$$\oint_{\Delta s} \mathbf{D} \cdot d\mathbf{s} = \int_v \rho_v \, dv$$



$$\boxed{\mathbf{n} \cdot (\mathbf{D}_1 - \mathbf{D}_2) = \rho_s \qquad \text{C/m}^2}$$

Normal Components of Electric Field

$$\boxed{\mathbf{n} \cdot (\mathbf{D}_1 - \mathbf{D}_2) = \rho_s \qquad \text{C/m}^2}$$

Special Case: Two Perfect Dielectrics

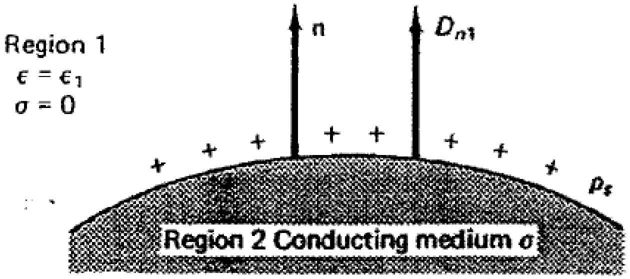
$$\mathbf{n} \cdot (\mathbf{D}_1 - \mathbf{D}_2) = 0$$

$$\boxed{D_{n1} = D_{n2}}$$

Special Case: Metal/Dielectric

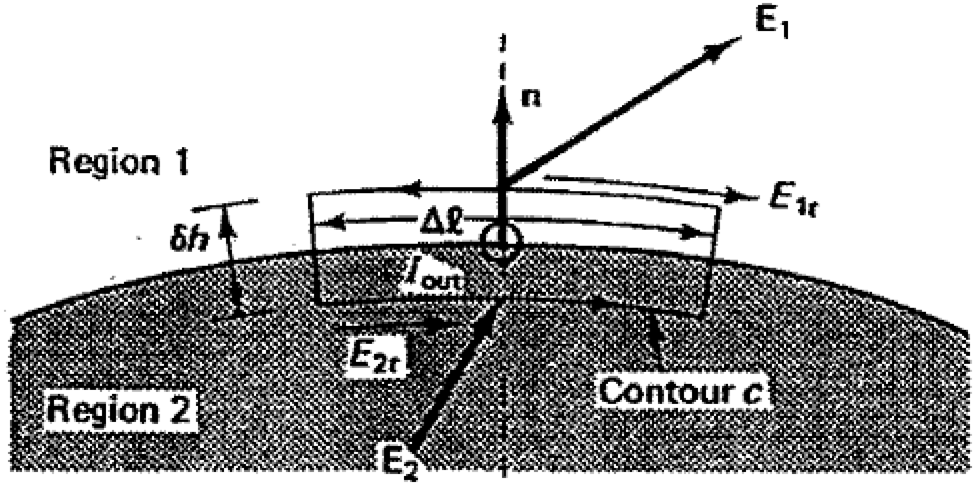
<u>For static fields</u>, the electric field inside the conductor is zero.

$$\boxed{\mathbf{n} \cdot \mathbf{D}_1 = \rho_s}$$



Tangential Component of Electric Field

$$\boxed{\oint_c \mathbf{E} \cdot d\ell = -\frac{d}{dt}\int_s \mathbf{B} \cdot ds}$$



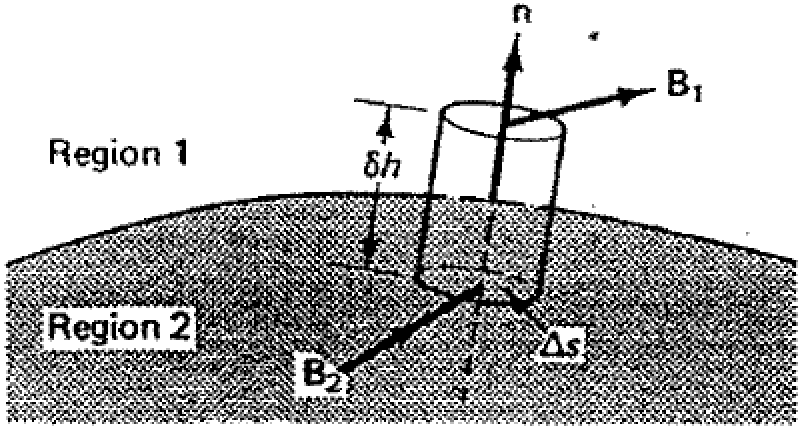
$$E_{2t}\Delta\ell - E_{1t}\Delta\ell = -\frac{d}{dt}(\mathbf{B} \cdot \Delta\ell\, \delta h\, \mathbf{a}_{out})$$

$$E_{2t} - E_{1t} = 0 \qquad E_{2t} = E_{1t}$$

$$\boxed{\mathbf{n} \times (\mathbf{E}_1 - \mathbf{E}_2) = 0}$$

Normal Component of Magnetic Field

$$\boxed{\oint_s \mathbf{B} \cdot d\mathbf{s} = 0}$$



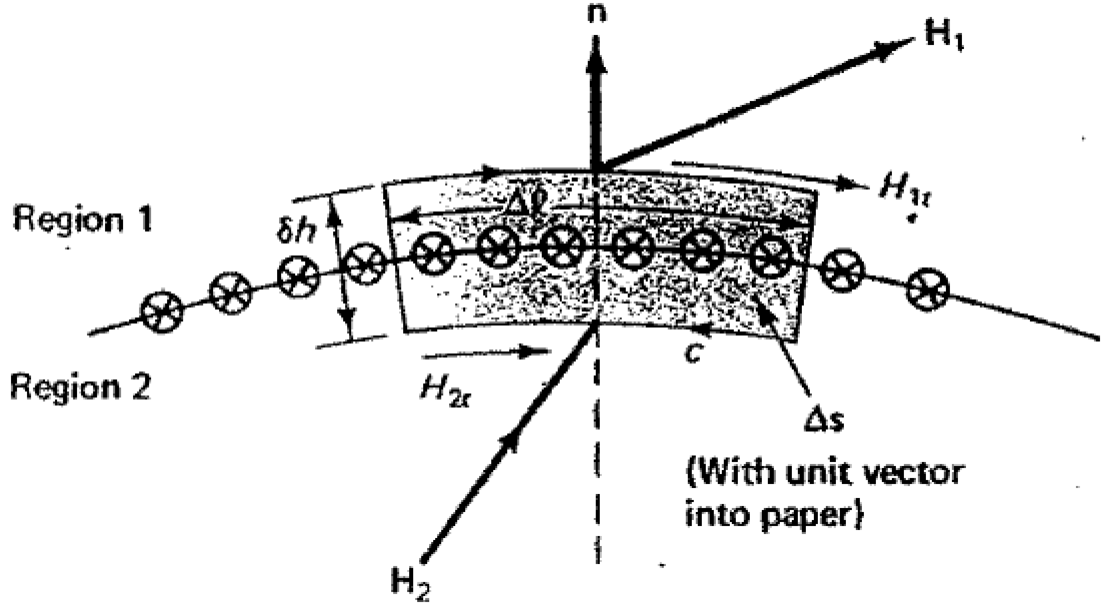
$$\oint_s \mathbf{B} \cdot d\mathbf{s} = B_{1n}\,\Delta s - B_{2n}\,\Delta s = 0$$

$$B_{1n} = B_{2n}$$

$$\boxed{\mathbf{n} \cdot (\mathbf{B}_1 - \mathbf{B}_2) = 0}$$

Tangential Component of Magnetic Field

$$\boxed{\oint_c \mathbf{H}\cdot d\ell = \int_s \mathbf{J}\cdot ds + \frac{d}{dt}\int_s \mathbf{D}\cdot ds}$$



$$H_{1t}\Delta\ell - H_{2t}\Delta\ell = \mathbf{J}\cdot\Delta\mathbf{s} + \frac{d}{dt}(\mathbf{D}\cdot\Delta\mathbf{s})$$

$$\lim_{\delta h\to 0} J_{in}\,\Delta\ell\,\delta h = J_{s(in)}\Delta\ell$$

$$H_{t1} - H_{t2} = J_{s(in)}$$

$$\boxed{\mathbf{n}\times(\mathbf{H}_1 - \mathbf{H}_2) = \mathbf{J}_s}$$

Summary of Boundary Condition

General case. Normal components of $\mathbf{D}$ are discontinuous by surface charge layer ρ_s, where ρ_s is a layer of *free charge* (bound charge, such as polarization, does not count for it has already been included in $\mathbf{D}$), hence

$$\boxed{\mathbf{n} \cdot (\mathbf{D}_1 - \mathbf{D}_2) = \rho_s} \tag{3.54}$$

Normal components of $\mathbf{B}$ are continuous because no magnetic charge exists, hence

$$\boxed{\mathbf{n} \cdot (\mathbf{B}_1 - \mathbf{B}_2) = 0} \tag{3.55}$$

Tangential components of $\mathbf{E}$ are always continuous at the interface between two media, hence

$$\boxed{\mathbf{n} \times (\mathbf{E}_1 - \mathbf{E}_2) = 0} \tag{3.56}$$

Tangential components of $\mathbf{H}$ are discontinuous by surface current density $\mathbf{J}_s$, hence

$$\boxed{\mathbf{n} \times (\mathbf{H}_1 - \mathbf{H}_2) = \mathbf{J}_s} \tag{3.57}$$

Summary

- Physical quantities to account for material properties:

  $\vec{E}$, $\vec{D}$, $\vec{J}$, $\vec{P}$ - vectors

  ρ, ϵ_r - scalars

  ϵ_0- constant

  $\vec{H}$, $\vec{B}$, $\overrightarrow{J_m}$, $\vec{M}$ - vectors

  χ_m, μ_r - scalars

- μ_0 - constant
- Material properties are accounted in Maxwell's equations via σ, χ_e, and χ_m.

  $\epsilon_r = 1 + \chi_e; \mu_r = 1 + \chi_m$

  $\epsilon = \epsilon_0 \epsilon_r$, and $\mu = \mu_0 \mu_r$

- Boundary conditions are derived from Maxwell's equations.